

Anticipation and Structural Coupling: Two Sides of the Same Coin

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Abstract

Two foundational frameworks in theoretical biology, namely Maturana's Structural Coupling and Rosen's Anticipatory Systems, developed in mutual isolation despite addressing the same phenomenon. We argue that Structural Coupling is precisely the historical process by which an organism becomes an anticipatory system: the process by which the regulatory subnetwork's dynamics come into conjugacy with relevant environmental dynamics. We show that the components of Rosen's anticipatory architecture map onto the autopoietic system in a way that satisfies all of Rosen's formal conditions. Furthermore, a conditional probability matrix describing the regulatory subnetwork is actually the observer's statistical re-description of a *bona fide* dynamical system. Because organisms co-constitute rather than merely predict their environments, Rosen's concordance condition must be reformulated as a fixed-point condition. This synthesis carries direct consequences for Artificial Life: current AI systems, including LLMs, are reactive in Rosen's precise sense and incapable of Umwelt construction; open-ended evolution requires true anticipatory agents. A criterion is provided.

Introduction

Living systems are unique in their autonomy, their capacity for self-fabrication, and their congruence with the environment that contains them. Our efforts to explain them fall into two unequal traditions: Experimental Biology, the undeniable motor of modern biology, and Theoretical Biology. Yet two centuries of observational data have not produced an understanding of biological *organization*.

Organization should be studied not as an *explanandum* but as an *explanans*: a *principle* for biology. Theoretical Biology, the tradition that should provide this, has remained a minor actor, and even recent work still asks for an adequate definition of *organization* (Mossio, 2024).

After World War II, the digital-computer metaphor for metabolism and the nervous system, alongside cybernetics, constrained biological thinking by privileging input–

output descriptions over organizational ones (Rosen, 1991). A sub-tradition then emerged, which disputed the Computer/Information metaphor for failing to reflect a basic condition of living systems: self-fabrication. Among those who advanced this interpretation (working largely in isolation) were Robert Rosen (1934–1998) with (M, R) –Systems, Humberto Maturana (1929–2021) & Francisco Varela (1946–2001) with Autopoietic Systems, and Tibor Gánti (1933–2009) with Chemoton Theory.

Maturana and Varela further interpreted living systems not as mere classifiers of reality but as *constructors of reality itself* (Maturana and Varela, 1980). This constructivist notion was termed *Structural Coupling* by Maturana, and *Enaction* by Varela, both extending von Uexküll's *Umwelt* (von Uexküll, 1909): the portion of the world that exists *for* an organism, whose objects are given only relative to its perception–action cycle, i.e. von Uexküll's *Funktionskreis*. Maturana and Varela radicalize this, the Umwelt being not merely perceived but enacted.

Maturana held that his main contribution was Structural Coupling, not autopoiesis (Maturana, 2012; Maturana et al., 2011): his first theoretical work concerned how the nervous system *constructs* reality (“Biology of Cognition,” Report #9, BCL, 1969), and autopoiesis was a translation of that idea to metabolism.

From the same sub-tradition, Robert Rosen had an independent intuition in the late 1950s (Rosen, 1958a,b, 1959): self-fabrication can be *logically deduced* from the connectivity of a network with catalytic closure, an entity he called an (M, R) –System. In his early writing (1958–1980) the “complete” metabolism is represented by a single symbol f (Letelier et al., 2003, 2006; Soto-Andrade et al., 2011). However, Rosen's discovery has faced severe obstacles in its translation to concrete biological examples (Letelier et al., 2006; Cárdenas et al., 2010; Gatherer and Galpin, 2013; Hofmeyr, 2021). In 1985, Rosen published “*Anticipatory Systems*” (second edition; Rosen, 2012), where he explored potential connections between self-fabrication and anticipatory behavior, a process he considers, without using the term, as necessarily *cognitive* as we shall explain below.

As Artificial Life reaches a turning point, with LLMs and possible open-ended evolution on the horizon (Karelin et al., 2025) and interest in Rosen’s Anticipatory Systems resurging (Astorga and Rosen, 2025), the moment is right to investigate the parallelism between Structural Coupling and Anticipation. By showing the interconnected contributions of Umwelt, Structural Coupling, and Anticipation, we advance a path forward amalgamating these models as the basis for a non-mechanistic¹ cognitive science in coherence with the Biology of Cognition.

Reactive Systems

As Rosen explored in diverse literature (Rosen, 1988, 1991, 2000), Western science made great progress on the shoulders of both the Cartesian assumption of fractionability and the Newtonian focus on recursion, which allows systems to be studied in terms of their states and parts. By a *system* we mean a state set together with a transition structure (Eqs. (1)–(2)); an *open system* also has input and output channels. System states $E(t)$ evolve according to

$$E(t+k) = H(E(t), p_1, \dots, p_n) \quad (1)$$

or in differential form:

$$dE(t) = F(E(t), t, p_1, \dots, p_n). \quad (2)$$

The future state $E(t+k)$ is thus computed from the present state $E(t)$, with H the discrete transition map, F the corresponding vector field, $p_1, \dots, p_n$ fixed parameters, and k a time-step. This is the world of feedback and reactive systems: the future of a system is merely some form of computation based on the system’s past.

A classical example is the cybernetic loop, where a system’s action on its environment is sensed as a feedback stimulus to which the system reacts.

Anticipatory Systems

According to Rosen (first edition (Rosen, 1985), second edition (2012)), an anticipatory system A is one which possesses a model M about itself and/or its environment which is used to proactively determine future states of A . In this definition, which *prima facie* does not appear radically different from the reactive paradigm, systems which are endowed with anticipation are assumed to be *feedforward* and *proactive* systems.

How can natural systems anticipate? Models here are neither passive nor perfect replicas: they are bounded, purposive, and built through interaction and trial-and-error. In Rosen’s terms (2012):

¹What we reject is the computational (and Cartesian) notion of mechanism. Although our account might appear compatible with New Mechanism (Machamer et al., 2000; Bechtel and Abrahamson, 2005; Craver, 2007), we agree with Daniel Nicholson and others (Nicholson, 2012) in that there might be no substance to New Mechanism at all.

As we recall, the decisive feature of an anticipatory system is that, in effect, it employs its model to make predictions about [a given natural system], and more importantly, change of state occurs in [the anticipatory system] at an instant as a function of its predictions. (p. 315)

The model captures only the regularities relevant to prediction. It is a *formal* counterpart compact enough to “simulate” the modeled system without requiring its nominal time: by running forward in a faster *surrogate* time, the system places itself in potential future states and selects an action **without inducing retrocausality**. That two times are needed points to the necessity of a theory of *Biological Time*.

Definition (anticipatory system). Let A and B be systems in the sense above. Then A is anticipatory if:

1. A contains a model M of another system B which does not exhaust B : A *interiorizes* **some** relevant meanings of B . In A there will exist a subsystem X containing the aspects of A which mediate the interaction with B and which represent the behaviors of A *if no model M existed*. M will not be influenced by X , but X will be influenced by M . X will change (pre-adapt) as M anticipates². Ergo, $A \rightarrow M \times X$. Some *subsystem* herein is a factor of the total state space; A encodes into $M \times X$, and $\hat{O}, \hat{S}, \sigma, \epsilon$ (introduced below) are likewise identified by the processes they comprise.
2. The present state of A at time t must encode in M a future state at time t' of B : the system is not merely reacting to actual states.
3. A must interact with B , using M ’s predictions of B ’s behavior, through an effector system E .
4. M preserves some relevant property of A , e. g.: preventing organizational dissipation.

Rosen’s original architecture places M , E , and B within the anticipatory A (Fig. 1). Input comes from the environment, and output into the same environment is produced by B . In reality, we must assume that X mediates between B and $M \times X$. For Rosen, a true anticipatory system is one in which the model and its predictions are functionally integrated into the system’s self-preserving organization: M is not “internal” in a representational sense but embodied in the system’s dynamics, whether distributed or localized (Astorga, 2025). This already hints at a confluence between autopoiesis and Rosennean anticipation.

For M to be predictive (condition 2), it must provide A with information regarding possible future states of B . If B evolves via a transformation T_t , then M must contain the

² M is *unlinked* to the observables coordinatizing X , but linked to their *rates of change*: each model state $m \in M$ fixes a copy of X whose dynamics depend on m (Rosen, 2012).

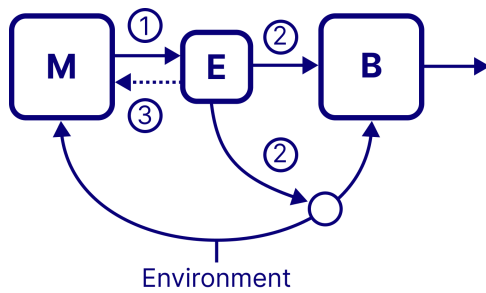
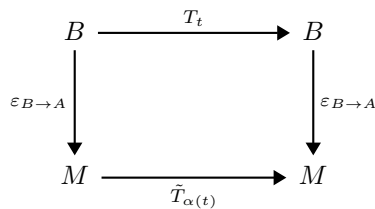


Figure 1: A classical Rosennean Anticipatory System where (3) allows the effector E to reset the model M and (2) allows it to change the dynamics of B by acting directly on it or on its environmental inputs based on M 's predictions passed on to E by (1).

corresponding transformation $\tilde{T}_{\alpha(t)}$ (where $\alpha(t) < t$ is the instant on the model's faster time scale corresponding to t) so that $\varepsilon_{B \rightarrow A}(T_t(B)) = \tilde{T}_{\alpha(t)}(\varepsilon_{B \rightarrow A}(B))$, which means the present state $s_t(M)$ encodes (ε) a future state $s_{t+i}(B)$. Hence, the evolution of $\tilde{T}_{\alpha(t)}$ must run faster than that of T_t (Rosen, 2012). This is the surrogate time referred to above, and as expressed diagrammatically:



This commuting diagram is what “ M models B ” means: a natural system B , a formal system M , and an encoding/decoding $\varepsilon_{B \rightarrow A}$ such that inference in M (via $\tilde{T}_{\alpha(t)}$) commutes with entailment in B (via T_t).

Rosen's ideas on Anticipation, even though developed by some authors (Dubois, 2003; Poli, 2017, 2019; Poli and Valerio, 2019), have mostly languished in obscurity, largely because of the dense mathematics and new intuitions demanded by Rosen's science³.

Structural Coupling

The history of Structural Coupling differs sharply from that of Rosen's Anticipation. Maturana's framework began with a simple internal report at Heinz von Foerster's Biological Computing Laboratory (BCL) in 1969. In “Report 9: *Biology of Cognition*,” Maturana proposed the following (Leterrier, 2022):

1. The nervous system does not function like a digital computer.

³Rosen never truly bridged his ideas on anticipation and organization represented by (M, R) -Systems. Rosen's *opus magnum*, “*Life Itself*” (Rosen, 1991), made little mention of Anticipation.

2. The stimulus–response metaphor is fundamentally flawed.
3. An organism's behavior is central to constructing its ecological niche.
4. “Objects” are not independent of the observer but are actively built as sensory–motor correlations.

Maturana and his student Francisco Varela applied idea (4) to cellular metabolism, developing the widely studied theory of Autopoietic Systems (Maturana, 1975). Varela's related formulation, known as Enaction, was introduced in *The Embodied Mind* (Varela et al., 1991), which paved the way for non-computational accounts of cognition.

To present Structural Coupling let us recover Maturana's and Varela's definition of autopoietic machines (Maturana & Varela, 1972):

An autopoietic machine is a machine organized as a system of processes of production of components concatenated in such a way that they produce components that i) generate the processes (relations) of production that produce them through their continuous interactions and transformations and ii) constitute the machine as a unity in physical space. Consequently, an autopoietic machine continuously specifies and produces its own organization through the production of its own components, under conditions of continuous perturbation and compensation of those perturbations (production of components). We can say then that an autopoietic machine is a homeostatic system that has its own organization as the variable that it maintains constant (p. 68, our translation).

Living Systems are thus *devices* that **act**, living in a loop (or spiral) connecting *Perception* to *Action*.

An Autopoietic network's structure changes constantly while its organization remains invariant. We distinguish two key internal subsets:

1. **Organization $\hat{O}$:** The subset of internal processes that define *class identity*, i.e., those that remain mostly invariant under diverse conditions.
2. **Structure $\hat{S}$:** Processes and components that may change without affecting class identity.

The invariance of $\hat{O}$ is definitional: $\hat{O}$ is the relational pattern whose persistence *is* the system's membership in its class, so while it holds the system remains that kind of thing and only $\hat{S}$ varies. Its boundary is therefore a viability threshold; crossing it is not a structural change but the loss of $\hat{O}$, i.e. death or a transition to another class. Within both sets, we can identify transversal subsets central to the theory:

1. **Sensorium σ :** subsets of Organization and Structure modulated by the biological function of *Perception*.
2. **Effectorium ϵ :** subsets of Organization and Structure modulated by the biological function of *Action*.

(In von Uexküll's terms, σ realizes the *Merkwelt* and ϵ the *Wirkwelt*; the loop they close is the *Funktionskreis*.) Many living systems have specialized sensoria/effectoria: birds can detect magnetic fields; bacteria act primarily by excreting chemicals. Thus, sensoria and effectoria must be understood as complex sensory-motor *geometries* where processes that connect the organism with its medium—its *Umwelt*—take place.

Internal Signals in Living Systems

When a circumstance triggers the sensorium, internal signals (molecular concentration vectors s_i) propagate through the system, interacting with components of the $\hat{O}$ and/or $\hat{S}$ to produce a contextually coherent action. Critically, there is no one-to-one mapping between external circumstances and the signals they evoke. Although living systems are historical, memory-bearing entities, they are not time-invariant: the same perturbations need not produce the same internal signals. Crucially, this historicity, as we will discuss later, is central to the Biology of Cognition.

Structural Coupling

Every living system (from *Yersinia pestis* to *Ornithorhynchus anatinus* to every member of genus *Homo*) is continuously immersed in a stream of perturbations that drive perception and action. The organism-environment relationship evolves through each successive (*perception* – *action*) loop, which also constantly changes $\hat{S}$ (see (Letelier et al., 2002), Fig. 2).

An example of Structural Coupling

Structural Coupling can seem mysterious (as if organisms construct their own environmental objects). A clarifying example is the bacterial production of atmospheric oxygen around 3,200 Mya.

When the first bacteria appeared (~3,850 Mya), the atmosphere contained no free oxygen: only water, nitrogen, methane, and carbon dioxide. For roughly 1,100 My, all life was anaerobic. Then ~3,200 Mya, one bacterial lineage evolved oxygenic photosynthesis, splitting water molecules to produce oxygen: initially a toxin lethal to competitors that had colonized every available anaerobic niche. Yet this same “poison” eventually drove the evolution of aerobic lineages that came to depend on it. Oxygen thus became a permanent feature of the world from ~2,480 Mya, at the cost of triggering the Huronian glaciation (2,400–2,100 Mya) and nearly extinguishing all life on Earth.

The story illustrates the core thesis: organism-environment interactions generate entirely new environmental features, including the selective pressures that will shape subsequent evolution.

Having presented both frameworks independently, we now turn to their integration.

Anticipation and Structural Coupling

We contend that Structural Coupling is the *historical process* by which an organism becomes an anticipatory system⁴. We develop this thesis below, but must first confront a difficulty that lies at the intersection of both frameworks: the status of *time*. According to Maturana (and von Uexküll before him), every dimension of our sensoriality derives from Structural Coupling (Maturana, 2002). Time is not a simple substance that flows homogeneously and regularly for all observers; it is itself a relational construct, *enacted* through the organism's history of interactions. Anticipation thus demands a temporal *mechanism*, even if the regularities it anticipates are spatial or otherwise structured rather than temporal. Rosennean Anticipation, moreover, requires not one but two flows of time: the nominal time of the environment and the surrogate time of the predictive model. As we shall argue, Structural Coupling provides the framework within which both temporalities can be grounded in *the concrete dynamics* of the organism.

The ballistic character of action

Every organismic action, however simple, has a temporal microstructure. When an action a is initiated, what it entails is a particular sequence of micro-events ($a = b_1 \rightarrow b_2 \rightarrow \dots \rightarrow b_n$). This sequence, which can be understood as a recursive chronicle, is essentially ballistic: once b_1 is initiated, the subsequent steps must unfold to completion. Consequently, the execution of b_j as a constituent of a (not the micro-event in isolation, which could belong to a different action) already *causally entails* both (I) the future execution of b_{j+1} and (II) the sensory state concomitant with b_{j+1} . The ballistic character alone is not anticipation: any deterministic chain, a falling rock included, entails its own future states. What makes the action anticipatory is that *which* sequence is initiated depends on the regulatory subnetwork, whose dynamics have already run ahead of the medium; that action-context, not the bare current state, carries the entailment.

The internal processes that sustain this ballistic trajectory (i.e., signaling cascades, allosteric modulations, regulatory dynamics) constitute a system whose dynamics unfold faster than the corresponding behavioral changes in the medium. Intracellular signaling operates in milliseconds to seconds

⁴Specifically: the process by which the regulatory subnetwork's dynamics come into *conjugacy* with the relevant environmental dynamics, thereby establishing the encoding that Rosen's framework requires.

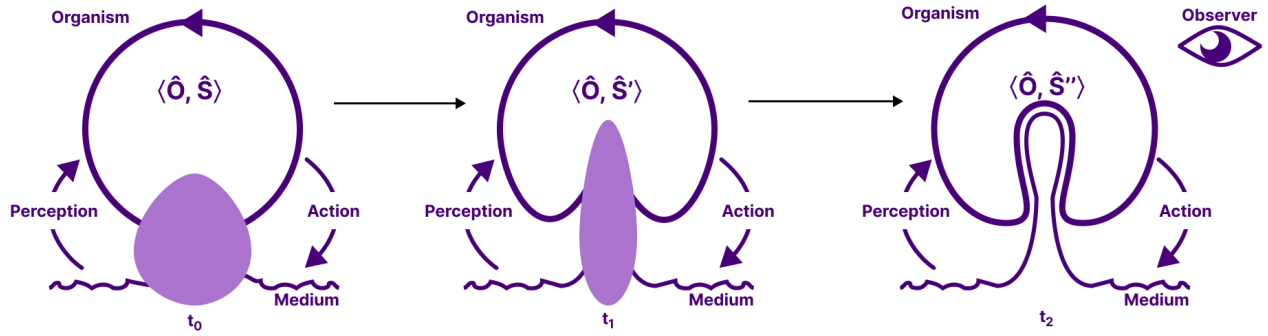


Figure 2: Mechanism of Structural Coupling: At t_0 the autopoietic system, with a given organization $\hat{O}$ and structure $\hat{S}$, confronts a new medium: relevant senso-motor correlations are not established, yet create a diffuse relationship (purple blob). By t_1 , new senso-motor correlations are created and the relationship is less diffuse (smaller purple blob). $\hat{O}$ remains invariant, but the $\hat{S}$ has changed to $\hat{S}'$. At t_2 , the relationship is clear: no blob, $\hat{O}$ remains invariant but $\hat{S}'$ has changed further to $\hat{S}''$, and the medium presents complementary changes. Naïve external observers erroneously describe the overall process as an *adaptation*.

and gene regulatory responses in minutes; whereas the environmental regularities to which the organism is coupled change in timescales of minutes to hours or longer. This timescale separation is a physical fact. The regulatory subnetwork that coordinates the unfolding of b_1 through b_n is already, in a precise sense, *running ahead* of the medium.

Three levels of internal signaling

We can now develop the concept of *internal signals* introduced above through three levels of description. These levels characterize the three stages in the refinement of the observer's account. The organism always operates as described at Level 2; it is the observer who, in attempting to understand the organism, may begin at Level 0 and progressively refine their description:

Level 0. At the simplest level, we describe the organism as possessing its invariance network $\hat{O}$ and a sensorium that receives signals s_α from the environment. The organism maintains associations between signals and actions: given signal s_α , some action X , Y , or Z is performed with certain regularity. The milieu presents distinct patterns $[X]$, $[Y]$, $[Z]$, which are environmental regularities the organism comes to treat as “objects”. But note: in this description, these objects *preexist*. They are assumed to be features of a given environment that the organism merely learns to classify. This is the standard cybernetic picture.

Level 1. At a more refined level, the associations between signals and actions can be described as a **Conditional Probability Matrix** (CPM): a table with signals as columns and actions as rows, whose entries $P(Y | s_\alpha)$, normalized so that $\sum_Y P(Y | s_\alpha) = 1$, give the regularities the observer sees in the organism's behavior. This CPM admits an **updating mechanism**: when s_α is produced, action Y is performed,

and organization is conserved, the corresponding entry is reinforced and its column renormalized, a Hebbian-like rule

$$P(Y | s_\alpha)_{t+1} = \frac{P(Y | s_\alpha)_t + \eta}{1 + \eta}, \quad \eta > 0, \quad (3)$$

with learning rate η and the remaining entries of the column rescaled by $1/(1 + \eta)$. Two things must be emphasized: First, the CPM is the *observer's* statistical description of the effective behavior of the organism's regulatory dynamics. This is consistent with Maturana's insistence that notions of coding, information, and regulation belong to the observer's domain, not to the system's (Rosen had the same intuitions, which he expressed succinctly in Rosen, 2012). Second, whatever the CPM describes is realized by concrete metabolic and signaling processes within $\hat{S}(t)$ (for a detailed description see Valdés-Zorrilla, 2024).

Yet, level 1 still assumes pre-existing objects $[X]$, $[Y]$, $[Z]$ as sources of distinct signals. The CPM account maps signals to actions, but the entities that produce the signals are taken for granted.

Level 2. In this level, objects $[X]$, $[Y]$, $[Z]$ do not pre-exist. They are built and continuously maintained by calibrating internal parameters through recurrent interaction. Each “object” is the limit of a trajectory in the $\langle \text{perception}, \text{action} \rangle$ phase space: a recurrent pattern of sensorimotor interaction that stabilizes through the organism's history of coupling. Different trajectories converge to different invariants; these invariants *are* what the organism treats as objects, and also are the furnishings of the organism's *Umwelt*. Level 2 is, in dynamical terms, the construction of an Umwelt, and the fixed-point condition below states when a stable one exists. At this level, the observer's description finally matches what the organism has

been doing all along: co-constituting the regularities it will encounter. This is precisely what an anticipatory system does.

The identification

In Rosen's original architecture, the adaptive anticipatory system can be identified with the organism. This architecture can be redrawn as a (*perception–action*), allowing for the overlaying of the elements of an autopoietic system on the Rosennean diagram of anticipation. To this end, we produce the following identification which is shown diagrammatically in figure 3:

1. **Model** M is realized by the regulatory and signaling subnetwork of the autopoietic system. This is the dynamical system whose faster-than-medium ballistic timescale makes anticipation possible. At Level 1, the external observer describes its effective behavior as the CPM; but M itself is not a probability table, it is a *bona fide* dynamical system with its own internal entailment $k(M)$, as Louie (2012) requires.
2. **Effectors** E correspond to the effectorium ϵ .
3. X corresponds to the current structural state $\hat{S}(t)$.
4. B is the medium (or something in the medium, like another system or the system modeling itself⁵).
5. **Encoding** $\varepsilon_{B \rightarrow A}$ is realized by the sensorium σ .
6. The **property** P that the anticipatory system maintains is the conservation of $\hat{O}$.
7. The **updating of** M **by** E is realized by the structural modifications that the effectorium's activity induces in the regulatory subnetwork (changes in enzyme concentrations, receptor densities, allosteric states, and gene expression profiles).

Conditions previously stated for anticipation are also satisfied:

- Condition 1 (A contains a model M of B) holds via the regulatory subnetwork's conjugacy with the medium through the sensorium encoding. Since that subnetwork captures only the (*perception–action*) mapping, A does not exhaust B : the rest of the metabolic network constitutes X . The selected action modulates the effectorium and thereby alters structural degrees of freedom beyond M , so X is changed by M .

⁵Rosen permits M to model the system itself, its environment, or both. In the autopoietic case this disjunction is not resolved but dissolved: the medium that M models is partly constituted by the actions M selects, so self-modeling and world-modeling are not separable.

- Condition 2 (M is predictive) is satisfied by the physical timescale separation between intracellular signaling and environmental change, as argued above.
- Condition 3 (A must interact with B using M 's predictions) is satisfied because the effectorium's output is selected by M 's faster-timescale, not by the current sensory state. The interaction with B is therefore mediated by M 's predictions.
- Condition 4 (M preserves some property) is satisfied because M selects the action before some perturbation propagates to the organizational core, and $\hat{O}$ is conserved proactively rather than by reactive repair. The same timescale separation that grounds Condition 2 also grounds Condition 4.

Our contention is that this identification preserves the Rosennean structure: As Louie (2012) emphasizes, the predictive model in an anticipatory system must not be associated with any kind of probabilistic certainty about the future. Rosen's theory is a general qualitative theory about entailment structures. The regulatory subnetwork satisfies this: its dynamics constitute a genuine inferential entailment i within the organism, conjugate (via the sensorium encoding) to the causal entailment c in the milieu.

There is, however, a critical point of divergence. In Rosen's anticipation, the "environment" is not "*Umwelted*": it exists by itself, with its own dynamics independent of the anticipatory system. The concordance condition (Eq. 6.21 in (Rosen, 2012)) compares the model's prediction against the *actual* future state of an independently evolving system (p. 328):

$$\vec{u}_p(t+h) = \vec{u}(t+h) \quad (4)$$

In Structural Coupling, the organism's action at time t partially determines the medium's configuration at $t+h$. The organism does not predict an independently evolving world; *it co-constitutes the regularity it anticipates*.

This requires a reformulation of the concordance condition. Let μ denote the function describing how the organism's action transforms the medium, σ the sensorium mapping, and π the action selected by the regulatory subnetwork. The organism-medium loop is defined by

$$s(t+h) = \sigma(\mu(\pi(s(t)), m(t))). \quad (5)$$

Since the medium M models is partly constituted by the actions M selects, forcing a choice between $B = \hat{S}(t)$ and $B = \text{medium}$ imposes a separation that coupling dissolves; the autopoietic case generalizes Rosen's architecture rather than instancing it. Coupling is achieved when this loop has stable recurrent patterns: when M selects actions whose resulting medium is consistent with M 's own entailments. Rather than a matching condition between prediction and

an independently given reality, we have a *fixed-point condition*: the model must select actions that produce a world compatible with its own dynamics, subject to the constraint that $\hat{O}$ is conserved. The mathematical object we commit to is therefore the conjugacy of entailment structures given by the commuting diagram above, together with this fixed-point condition. The existence, uniqueness, and stability of such fixed points are open mathematical questions; the sheer fact that organisms achieve and maintain structural coupling serves as an existence proof. This fixed-point condition constitutes the core of the formal research program implied by the synthesis we have proposed.

This captures what each framework lacks: against Rosen, the organism is not merely a predictor but a *co-author* of the regularity it predicts; against Maturana, that co-authorship runs through a specific mechanism, the faster-than-nominal-time dynamics of the regulatory subnetwork.

Still, the medium is open: it is subject to perturbations not encoded in the regulatory subnetwork. As the medium changes beyond the scope of the model, adequacy of M progressively fails. This is Rosen's temporal spanning: the interval during which property P is maintained is finite because the model is necessarily an incomplete closed system encoding an open system (Rosen, 2012).

In autopoiesis the equivalent is the loss of structural coupling: the organism's structure no longer admits the medium's perturbations, not because a component has broken but because its structural trajectory has diverged from the medium's. This is a *global failure*: what fails is the *relational conjugacy* between model and environment, not any part.

Unlike Rosen's treatment, where temporal spanning leads inexorably to system failure, the Structural Coupling framework admits a dynamic of *re-coupling*. If $\hat{O}$ is not lost, the regulatory subnetwork can undergo further structural modifications that restore commutativity. This is what an external observer would describe as *learning*. Learning is re-coupling as re-anticipation, a resource that Rosen's framework, focused on the degradation of models, does not naturally provide.

That the fixed point is so often *lost* (in extinct lineages, and arguably in our own species' relation to its niche) is a consequence of the same openness⁶.

Strong & Weak Anticipation

Structural Coupling also bridges the gap between weak and strong modes of anticipation as distinguished by Dubois (2003). Weak anticipation employs a decoupled internal model, whereas strong anticipation demands a dynamical structure within the metabolism itself, isomorphic to the

⁶A lineage that perfectly and permanently satisfied the fixed point in a closed world could not evolve. Whether evolution ought to safeguard coupling more reliably is a normative question beyond our scope.

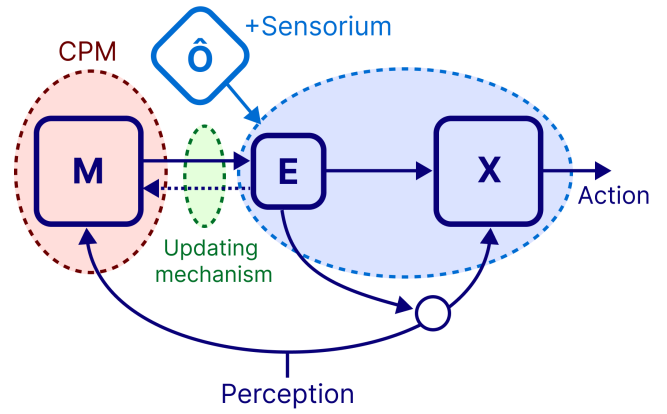


Figure 3: The diagram of an anticipatory system with autopoietic aspects superimposed. The corresponding object-system B is replaced by internal X .

organism-medium interaction (in a mechanistic sense: a real analogue computer).

The mechanisms of Structural Coupling correspond to strong anticipation precisely because the regulatory subnetwork functioning as M is not a detachable computational component but an integral part of $\hat{S}(t)$ whose dynamics are metabolic processes with a faster timescale arising from the physical properties of $\hat{S}(t)$'s molecular constituents. The conjugacy between model and environment is achieved not by engineering a separate representation but by the history of structural coupling itself reshaping the regulatory network until its internal entailment tracks the relevant environmental entailment.

Relation to the Free Energy Principle

Our fixed-point condition resembles active inference under the Free Energy Principle (FEP), where an agent updates a model and acts to make the world conform to its predictions (Friston, 2010, 2013); the related internal model principle has recently received Bayesian (Baltieri et al., 2025) and good-regulator (Virgo et al., 2025) formulations, with Baltieri and Suzuki (2025) relating these to Rosen. Active inference is nonetheless not an instance of our framework. It is probabilistic, since free-energy minimization is Bayesian inference, whereas Rosennean anticipation is a qualitative theory of entailment with nothing to do with probability (Louie, 2012). And where the good regulator's model is imposed on the system by an observer, our Level 1 already assigns that observer-relative, statistical description (the CPM) to the observer's domain; the organism's M at Level 2 is a *bona fide* dynamical system with its own entailment, not a belief state⁷.

⁷*Nota bene*: the FEP presupposes a non-equilibrium steady state, hence a time-invariance at odds with the historicity of coupling; it identifies the system with a Markov blanket rather than

Nova Synthesis

Anticipation theory introduces conceptual dimensions not explicitly articulated in Structural Coupling. Conversely, Structural Coupling foregrounds mechanisms of object and world-constitution absent from Rosen's treatment, where the environment is taken as independently given. Moreover, Structural Coupling provides what Rosen's framework lacks: a historical account of how the encoding functor is established, maintained, and restored after degradation, along with a strong account of cognition.

Since the 1920s many have held that something fundamental is missing from our frameworks for organisms: a theory of organization for Rashevsky, of self-fabrication for Rosen, Maturana, Varela, and Gánti.

We hypothesize that a comprehensive theory of biological organization can be achieved through a *novel synthesis* of existing frameworks. Two steps are critical. First, Structural Coupling and Anticipation must be integrated, as together they address niche construction more comprehensively than Umwelt as originally introduced. This integrated framework provides the proper foundations for a Biology of Cognition urgently needed for contemporary inquiries—such as in multi-species systems like holobionts, where recent work on entangled agencies (Sepúlveda-Pedro et al., 2026) has highlighted the need to account for individual-level agency within collective sympoietic dynamics.

Second, any rigorous theory of biological organization must begin with these foundations. Cognition is not an emergent feature of autopoietic networks but a condition of their very existence: anticipation is built into the regulatory dynamics that enable self-fabrication. Thus, cognition and biological organization mutually presuppose each other and are co-extensive.

Conclusions

The *nova synthesis* has concrete consequences for Artificial Life. Contemporary AI (LLMs, deep reinforcement learning, generative networks), however sophisticated, is reactive in Rosen's sense: it operates within the paradigm of Eqs (1)–(2), computing future states from past ones via fixed or gradient-updated parameters. They do not contain internal predictive models whose dynamics run on a faster timescale; they contain statistical compressions of training data⁸. As Louie (2012) emphasizes, Rosen's theory of anticipation has nothing to do with probability. A conditional probability distribution over next tokens, however large and however useful, is not a predictive model in the Rosennean sense: it is a

operational closure; and it sets aside the organization/structure distinction on which our account turns (Di Paolo et al., 2022).

⁸These are computational dynamics over token sequences, thus not conjugate to environmental dynamics via an encoding functor on a faster timescale: no computational system has dynamics that are *physically conjugate* to environmental dynamics, because computation is substrate-neutral.

summary of past correlations, not a dynamical system with its own internal entailment running ahead of reality.

Simple systems do not make mistakes: an error presupposes an internal norm against which behavior can be judged inadequate. In anticipation that norm is P , so when M selects an action that fails to preserve $\hat{O}$ a genuine organizational error occurs. Reactive systems have no such norm; its absence is structural. Rosen's result on the uncomputability of organisms has been contested, but analysis under the Sieg–Gandy account of computability suggests the processes constitutive of genuine anticipatory organization lie outside what any computing machine can realize (Astorga, 2025).

It is worth separating two claims often run together. The *architectural* claim is that current models lack faster-than-medium internal dynamics conjugate to the environment; this is contingent and could in principle be re-engineered. The *constitutive* claim, which we endorse, is stronger: anticipation requires closure to efficient causation (Louie, 2012), and cognition and biological organization are co-extensive (Froese and Ziemke, 2009), so a digital system could anticipate only by genuinely self-fabricating. Given the substrate-neutrality of computation noted above, no purely computational system meets this condition.

Indeed, a reactive system cannot co-constitute its environment; it only responds to inputs. An anticipatory system acts on its model's predictions and thereby transforms the medium it next encounters. The fixed-point condition (5) is what no reactive system achieves: the generation of stable sensorimotor invariants (objects, niches, *Umwelten*) by the organism's own action. Current artificial systems classify a given world; they do not construct one.

This bears on open-ended evolution, for which LLMs are now proposed as candidates (Karelin et al., 2025). In most artificial platforms the environment, and thus the fitness landscape, is given independently of the agents; it is not *Umwelted*. If organisms are *Umwelt* constructors, the landscape is instead continuously reshaped by the agents adapting to it: niche construction as a constitutive feature, not a side effect. Open-ended evolution therefore requires agents whose models and environments are coupled in a fixed-point relation; without it, everything behaves like a simple system and evolution converges.

This Level progression defines a criterion: Level 0 artificial evolving systems cannot achieve open-ended dynamics. Level 1 systems may exhibit longer transients but will eventually exhaust the novelty of a fixed environment. Only Level 2 systems, where the environment is a co-product, have the structural prerequisites for genuinely open-ended dynamics.

An artificial life system worthy of its name must both self-fabricate and *Umwelt*-construct. Up until now, an LLM's *Umwelt* is not self-constructed but derived from others' *Umwelten*, leaving it without any clear biological congruity.

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References

- Astorga, E. F. S. (2025). Of organisms and machines: Robert Rosen's mechanisms and machines under the light of siegandy computability. Master's thesis, National Autonomous University of Mexico, CDMX, Mexico.
- Astorga, E. F. S. and Rosen, J. (2025). International Conference on Anticipatory Systems and Rosennean Complexity (ICAS) @ UNAM-MX - Book of Abstracts.
- Baltieri, M., Biehl, M., Capucci, M., and Virgo, N. (2025). A Bayesian interpretation of the internal model principle. arXiv:2503.00511 [math.OC]. Preprint.
- Baltieri, M. and Suzuki, K. (2025). Mathematical approaches to the study of agents. PsyArXiv. Preprint; in press, Philosophical Transactions of the Royal Society B.
- Bechtel, W. and Abrahamsen, A. (2005). Explanation: A mechanist alternative. *Studies in History and Philosophy of Biological and Biomedical Sciences*, 36(2):421–441.
- Cárdenas, M. L., Letelier, J.-C., Gutiérrez, C., Cornish-Bowden, A., and Soto-Andrade, J. (2010). Closure to efficient causation, computability and artificial life. *Journal of Theoretical Biology*, 263(1):79–92.
- Craver, C. F. (2007). *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford University Press.
- Di Paolo, E. A., Thompson, E., and Beer, R. D. (2022). Laying down a forking path: Tensions between enaction and the free energy principle. *Philosophy and the Mind Sciences*, 3:2.
- Dubois, D. M. (2003). Mathematical Foundations of Discrete and Functional Systems with Strong and Weak Anticipations. In Goos, G., Hartmanis, J., Van Leeuwen, J., Butz, M. V., Sigaud, O., and Gérard, P., editors, *Anticipatory Behavior in Adaptive Learning Systems*, volume 2684, pages 110–132. Springer Berlin Heidelberg, Berlin, Heidelberg. Series Title: Lecture Notes in Computer Science.
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.
- Friston, K. (2013). Life as we know it. *Journal of the Royal Society Interface*, 10(86):20130475.
- Froese, T. and Ziemke, T. (2009). Enactive artificial intelligence: Investigating the systemic organization of life and mind. *Artificial Intelligence*, 173(3–4):466–500.
- Gatherer, D. and Galpin, V. (2013). Rosen's (M,R) system in process algebra. *BMC Systems Biology*, 7:128.
- Hofmeyr, J.-H. S. (2021). A biochemically-realizable relational model of the self-manufacturing cell. *Biosystems*, 207:104463.
- Karelin, G., Nakajima, K., Soto-Astorga, E. F., Carr, E., James, M., and Froese, T. (2025). Indeterminism in large language models: An unintentional step toward open-ended intelligence. Submitted to Artificial Life journal.
- Letelier, J. (2022). Cybernetics in Chile: a history with unexpected chapters. *AI and SOCIETY*, 37:1105–1113.
- Letelier, J. C., Marín, G., and Mpodozis, J. (2002). *Computing with Autopoietic Systems*, pages 67–80. Springer London, London.
- Letelier, J.-C., Marín, G., and Mpodozis, J. (2003). Autopoietic and (M,R) systems. *Journal of Theoretical Biology*, 222:261–272.
- Letelier, J.-C., Soto-Andrade, J., Guíñez Abarzúa, F., Cornish-Bowden, A., and Cárdenas, M. L. (2006). Organizational invariance and metabolic closure: Analysis in terms of (M,R) systems. *Journal of Theoretical Biology*, 238(4):949–961.
- Louie, A. H. (2012). Anticipation in (M,R)-systems. *International Journal of General Systems*, 41(1):5–22.
- Machamer, P., Darden, L., and Craver, C. F. (2000). Thinking about mechanisms. *Philosophy of Science*, 67(1):1–25.
- Maturana, H. (2002). Autopoiesis, structural coupling and cognition: A history of these and other notions in the biology of cognition. *Cybernetics Human Knowing*, 9(3-4):5–34.
- Maturana, H. (2012). Reflections on my collaboration with Francisco Varela. *Constructivist Foundations*, 7(3):155–164.
- Maturana, H., Paucar-Caceres, A., and Harnden, R. (2011). Origins and implications of autopoiesis. preface to the second edition of *De Máquinas y Seres Vivos*. *Constructivist Foundations*, 6(3):293–306.
- Maturana, H. R. (1975). The organization of the living: A theory of the living organization. *International Journal of Man-Machine Studies*, 7(3):313–332.
- Maturana, H. R. and Varela, F. J. (1980). *Autopoiesis and Cognition: The Realization of the Living*, volume 42 of *Boston Studies in the Philosophy of Science*. D. Reidel, Dordrecht.
- Mossio, M. (2024). Introduction: Organization as a Scientific Blind Spot. In Mossio, M., editor, *Organization in Biology*, volume 33 of *History, Philosophy and Theory of the Life Sciences*, pages 1–22. Springer International Publishing, Cham.
- Nicholson, D. J. (2012). The concept of mechanism in biology. *Studies in History and Philosophy of Biological and Biomedical Sciences*, 43(1):152–163.
- Poli, R. (2017). *Introduction to Anticipation Studies*, volume 1 of *Anticipation Science*. Springer International Publishing, Cham.
- Poli, R., editor (2019). *Handbook of Anticipation: Theoretical and Applied Aspects of the Use of Future in Decision Making*. Springer International Publishing, Cham.

- Poli, R. and Valerio, M., editors (2019). *Anticipation, Agency and Complexity*, volume 4 of *Anticipation Science*. Springer International Publishing, Cham.
- Rosen, R. (1958a). A relational theory of biological systems. *Bulletin of Mathematical Biophysics*, 20:245–260.
- Rosen, R. (1958b). The representation of biological systems from the standpoint of the theory of categories. *Bulletin of Mathematical Biophysics*, 20:317–341.
- Rosen, R. (1959). A relational theory of biological systems II. *Bulletin of Mathematical Biophysics*, 21:109–128.
- Rosen, R. (1985). *International federation for systems research*. Elsevier.
- Rosen, R. (1988). Effective processes and natural law. In Herken, R., editor, *The universal Turing machine: a half century survey*, Oxford science publications, pages 485 – 498. Oxford Univ. Press, Oxford.
- Rosen, R. (1991). *Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life*. Columbia University Press, New York.
- Rosen, R. (2000). *Essays on Life Itself*. Columbia University Press, New York.
- Rosen, R. (2012). *Anticipatory Systems: Philosophical, Mathematical, and Methodological Foundations*, volume 1 of *IFSR International Series in Systems Science and Systems Engineering*. Springer New York, New York, NY. Original publish date: 1985.
- Sepúlveda-Pedro, M. A., Ramírez-Vizcaya, S., Hernández-Ochoa, J. L., Vergara-Silva, F., and Gastelum-Vargas, M. (2026). Entangled Agencies: The Enactive Approach and Eco-Evo-Devo in the Scientific Study of Sympoiesis as a Dialectics of Life. *Adaptive Behavior*, page 10597123251411884.
- Soto-Andrade, J., Jaramillo-Riveri, S., Gutiérrez, C., and Letelier, J.-C. (2011). Ouroboros avatars: A mathematical exploration of self-reference and metabolic closure. In *Proceedings of the 11th European Conference on Artificial Life (ECAL 2011)*, pages 763–770. MIT Press.
- Valdés-Zorrilla, A. (2024). A categorical approach to autopoietic structural coupling. Master’s thesis, Universidad de Chile, Santiago, Chile.
- Varela, F. J., Thompson, E., and Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press, Cambridge, MA.
- Virgo, N., Biehl, M., Baltieri, M., and Capucci, M. (2025). A “good regulator theorem” for embodied agents. In *ALIFE 2025: Ciphers of Life. Proceedings of the Artificial Life Conference 2025*. MIT Press. Article 46.
- von Uexküll, J. (1909). *Umwelt und Innenwelt der Tiere*. Julius Springer, Berlin.